UNIVERSIDADE FEDERAL DE PERNAMBUCO

Centro de Informática
Programa de Pós-Graduação em Desenvolvimento de Software Automotivo

SYSTEM REQUIREMENTS DOCUMENT: ENGINE START-STOP SYSTEM (ESS)

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1 Introduction

1.1 Purpose and Intended Audience

In the global automotive landscape, the pursuit of energy efficiency and the minimization of environmental impact have become a challenge to technical and commercial goals. One of the main issues occurs during a typical urban circuit, where vehicles spend a significant portion of their time idling (e.g., at traffic lights, jams, or intersections). During these moments, the engine consumes fuel and emits greenhouse gases unnecessarily, wasting energy and degrading air quality. This document specifies the system and software requirements for the Engine Start-Stop System (ESS), whose primary goal is to reduce fuel consumption and CO_2 emissions by automatically stopping and restarting the internal combustion engine under specific driving conditions.

It is intended for system engineers, software developers, and validation teams involved in the ESS development lifecycle, serving as the main reference for design decisions, implementation, and verification activities throughout the project.

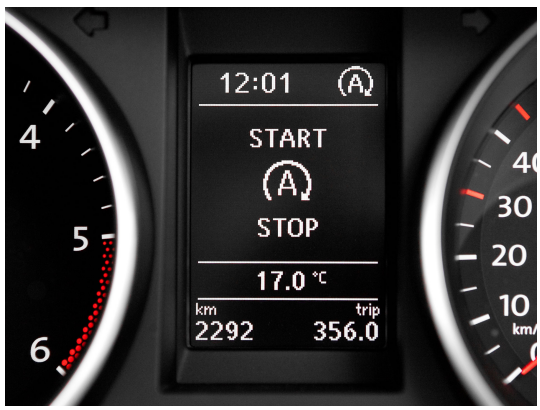


Figure 1: Start-Stop System

1.2 Document Conventions

Requirements are uniquely identified using the following hierarchical format:

- `SYS-HL-FR-xx`: System-level, High-Level Functional Requirement
- `SYS-LL-FR-xx`: System-level, Low-Level Functional Requirement
- `SW-HL-FR-xx`: Software-level, High-Level Functional Requirement
- `SW-LL-FR-xx`: Software-level, Low-Level Functional Requirement
- `SYS-HL-NFR-xx`: System-level, High-Level Non-Functional Requirement
- `SYS-LL-NFR-xx`: System-level, Low-Level Non-Functional Requirement
- `SW-HL-NFR-xx`: Software-level, High-Level Non-Functional Requirement
- `SW-LL-NFR-xx`: Software-level, Low-Level Non-Functional Requirement

Priorities are classified as **High**, **Medium**, or **Low** at the high-level requirements.

Each functional requirement includes a **Traceability** field linking it to its parent requirement at the level above, following the derivation chain:

$$\texttt{SYS-HL} \rightarrow \texttt{SYS-LL} \rightarrow \texttt{SW-HL} \rightarrow \texttt{SW-LL}$$

This ensures full vertical traceability from stakeholder needs down to software implementation, and supports verification activities by allowing each requirement to be traced back to its origin.

Non-functional requirements, however, are not decomposed in the same hierarchical manner. Instead, they are specified at the level where they apply and are mapped alongside the related functional requirements, providing constraints or quality attributes that influence system behavior.

1.3 Product Scope

The ESS automatically shuts down the engine when the vehicle is stationary under appropriate conditions and restarts it when propulsion is requested by the driver or when any safety/operational constraint requires it. The system also provides driver-facing controls including a dedicated Start/Stop button for enable/disable control, and CO_2 savings estimation displayed on the instrument cluster.

The system scope is limited to automatic transmission vehicles. The following sensors and signals are considered as inputs within the defined scope:

- **Speed Sensor** - monitors vehicle speed to determine whether the standstill threshold has been reached and to enable Autostop eligibility.
- **Brake Pedal Position Sensor** - detects whether the driver is actively pressing the brake, which is a primary enabling condition for Autostop and a restart trigger when released.
- **Accelerator Pedal Position Sensor** - detects driver propulsion demand, triggering an immediate engine restart when pressed during Autostop.
- **Ignition Status** - determines whether the vehicle is powered on, enabling the ESS to initialize and set SS_Enabled = TRUE at every ignition-on event.
- **Start/Stop Button** - allows the driver to manually enable or disable the Start/Stop function at any time during vehicle operation.
- **Transmission Gear Selector** - provides the current gear position, used to verify Drive engagement as an Autostop condition and as a restart trigger when moved out of Drive.
- **Inclinometer/Grade Sensor** - measures the vehicle pitch angle to inhibit Autostop on steep inclines where engine-off could compromise vehicle stability and braking safety.
- **Driver Door Status Sensor** - detects whether the driver door is open or not fully closed, inhibiting Autostop as a safety measure against unintended vehicle exit during engine-off.
- **Seatbelt Status Sensor** - verifies that the driver seatbelt is fastened, inhibiting Autostop when the seatbelt is not engaged as a safety precaution.

1.4 Document Overview

This document is structured as follows:

- **Section 1 - Introduction**: presents the purpose, intended audience, scope, conventions, and overview of the document.
- **Section 2 - Use Case Diagram**: describes the system actors and use cases, including trigger conditions, include and extend relationships, and the corresponding UML diagram.
- **Section 3 - Functional Block Diagram**: identifies all subsystems, sensors, and interfaces within the ESS scope, illustrated by the functional block diagram.
- **Section 4 - State Machine Diagram**: describes the primary operating states of the ESS, the transitions between them, and the conditions that trigger each transition.
- **Section 5 - Simplified Requirements Diagram**: presents the four-level requirements hierarchy and the traceability relationships between levels.
- **Section 6 - Functional Requirements**: specifies all functional requirements at system and software level, organized from high-level to low-level with full traceability.
- **Section 7 - Non-Functional Requirements**: defines performance, timing, and safety constraints at system and software level.
- **Section 8 - Requirement Traceability**: describes the tool used for requirements traceability and its integration with project development and maintenance.
- **Section 9 - Traceability Matrix**: provides the full traceability mapping from SYS-HL down to SW-LL across all requirement levels.
- **Section 10 - Appendix**: provides supplementary material that supports the main content of the project, including detailed use case specifications.
- **Section 11 - References**: lists the scientific articles and manufacturer documentation used as basis for the requirements elicitation.

2 Use Case Diagram

2.1 ESS Actors

- **Driver** - Primary actor who interacts with the system through the brake pedal, accelerator, gear selector, and Start/Stop button.
- **Start/Stop Module** - Internal system actor responsible for evaluating operating conditions and commanding engine stop and restart.
- **Ignition System** - External actor that triggers system initialization at every ignition-on event, resetting timers and restoring the default enabled state.

- **Powertrain ECU** - External actor that executes engine stop and restart commands and provides vehicle motion and gear data to the module.
- **Instrument Cluster** - External actor responsible for presenting system status to the driver, including enable state, unavailability warnings, Autostop indication, and CO_2 savings.

2.2 Use Cases

The main use cases of the Start/Stop system are listed below. Detailed specifications for each use case are provided in Appendix 10.

- **UC1 – Execute AutoStop** (see Appendix 10.1.1)
- **UC2 – Execute AutoStart** (see Appendix 10.1.2)
- **UC3 – Enable/Disable StartStop Button** (see Appendix 10.1.3)
- **UC4 – Display AUTOSTOP ACTIVE Indicator** (see Appendix 10.1.4)
- **UC5 – Estimate CO_2 Avoided** (see Appendix 10.1.5)
- **UC6 – Display CO_2 Avoided During AutoStop** (see Appendix 10.1.6)
- **UC7 – Evaluate AutoStop Eligibility** (see Appendix 10.1.7)
- **UC8 – Evaluate AutoStart Eligibility** (see Appendix 10.1.8)
- **UC9 – Generate EngineStopRequest** (see Appendix 10.1.9)
- **UC10 – Generate EngineRestartRequest** (see Appendix 10.1.10)
- **UC11 – Indicate StartStop Enabled State and Changes** (see Appendix 10.1.11)
- **UC12 – Initialize StartStop** (see Appendix 10.1.12)
- **UC13 – Initialize Time AutoStop** (see Appendix 10.1.13)

2.3 Diagram

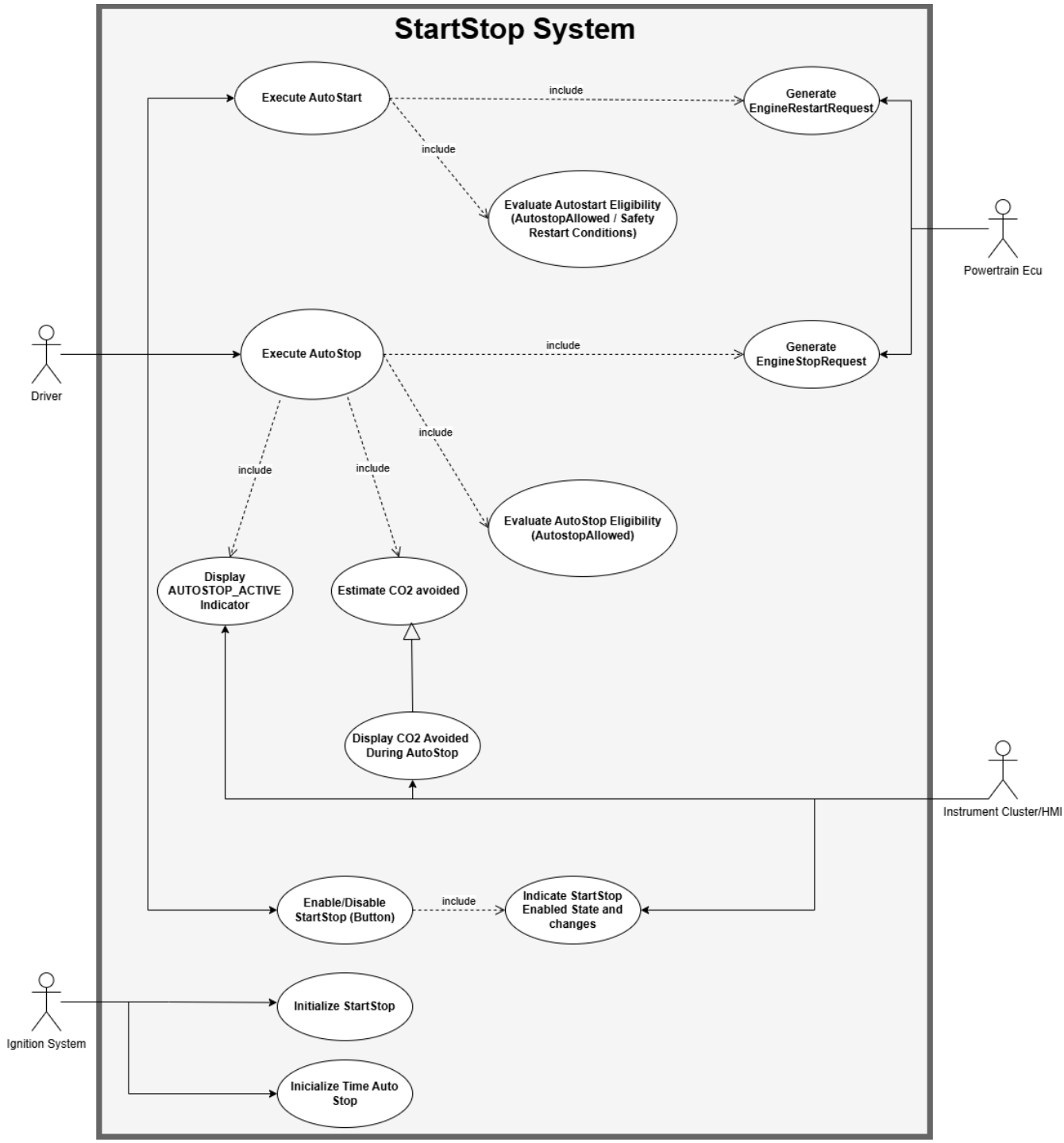


Figure 2: ESS Use Case Diagram

3 Functional Block Diagram

The ESS interacts with the following subsystems and sensors within the defined scope:

- **Powertrain ECU** - receives EngineStopRequest and EngineStartRequest commands from the Start/Stop Module and provides vehicle operational data such as VehicleSpeed, GearPosition, and IgnitionStatus.
- **Brake Pedal Sensor** - provides the BrakePedalPosition used to determine braking conditions required for AutoStop or restart events.
- **Accelerator Pedal Sensor** - provides the AcceleratorPedalPosition, used to detect driver intent to move the vehicle and trigger engine restart.
- **Speed Sensor** - provides VehicleSpeed, used to determine vehicle standstill conditions required for AutoStop activation.
- **Inclinometer/Grade Sensor** - provides the VehiclePitchAngle, allowing the module to inhibit AutoStop under excessive slope conditions.
- **Driver Door Sensor** - provides DoorStatus, used for safety monitoring and to prevent AutoStop under unsafe conditions.
- **Seatbelt Sensor** - provides SeatbeltStatus, supporting safety-related AutoStop inhibition logic.
- **Transmission Module** - provides GearPosition, used to determine whether the vehicle is in a valid driving state (e.g., Drive).
- **Start/Stop Button** - provides ButtonInput, allowing the driver to enable or disable the Start/Stop functionality.
- **Instrument Cluster** - receives status signals from the Start/Stop Module, including SSEnable, StopStatus, VehicleSpeed, CO2Avoided, and SSUnavailable, to inform the driver about system operation and environmental benefits.
- **Driver (User)** - interacts with the system by operating the pedals and the Start/Stop button and receives feedback through the instrument panel.

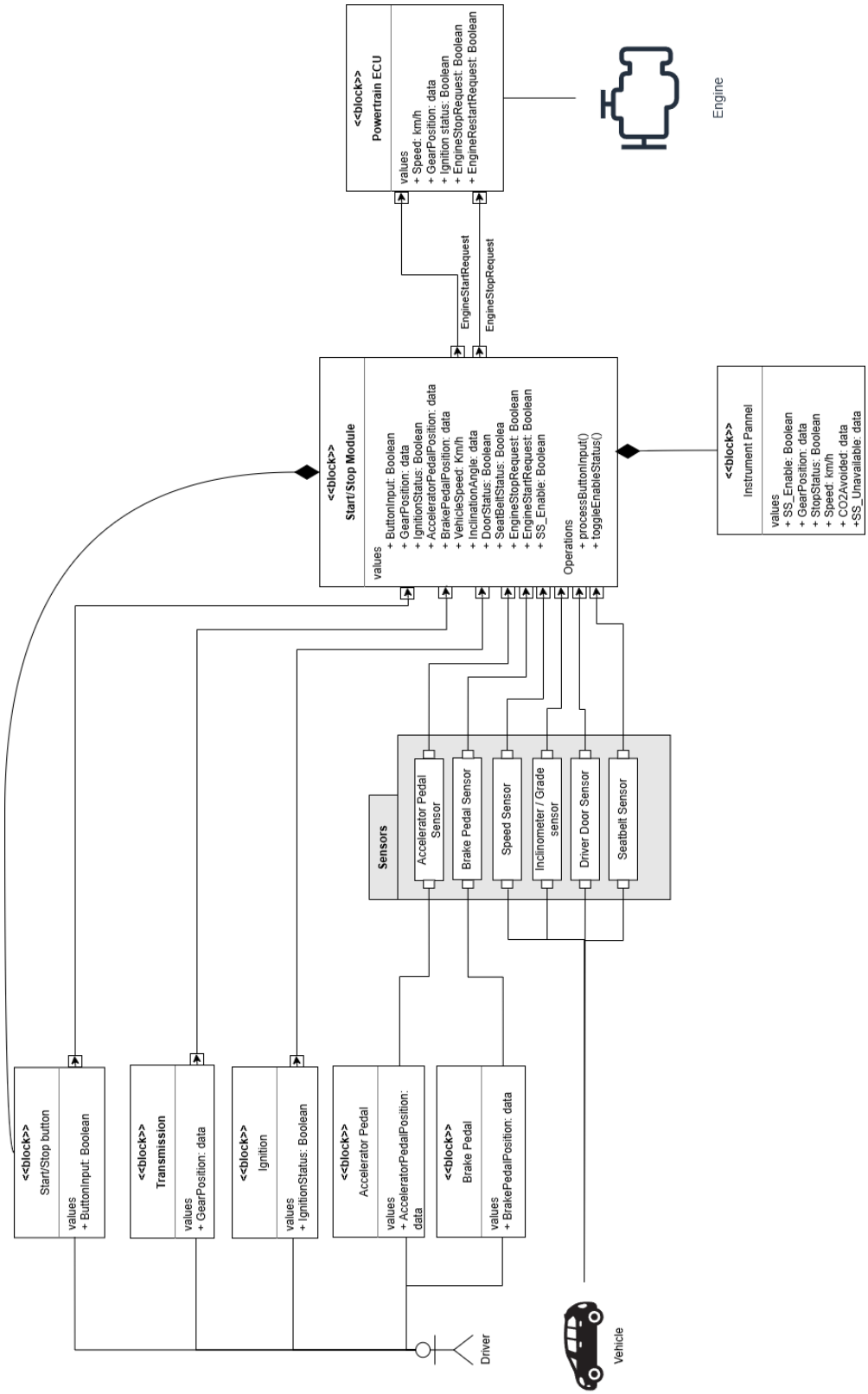


Figure 3: ESS Functional Block Diagram

4 State Machine Diagram

The ESS operates through the following primary states:

1. **Monitoring** - Default active state; Autostop eligibility is continuously evaluated each 100 ms cycle.
2. **Autostop** - Engine is stopped by ESS; CO_2 counting is active; Autostart conditions are monitored.
3. **Autostart** - Engine restart has been commanded; the system returns to Monitoring once the engine is running again.
4. **Unavailable** - A safety or operational inhibition is active (door open, seatbelt unfastened, inclination exceeded, etc.); Autostop is blocked automatically.
5. **Disabled** - Start/Stop function disabled by the driver via button press; no Autostop or Autostart logic is active.

States 1–4 are grouped inside the *System Active* super-state. The system transitions to *Disabled* from any active sub-state when the driver presses the button, and returns to *System Active* on a subsequent button press.

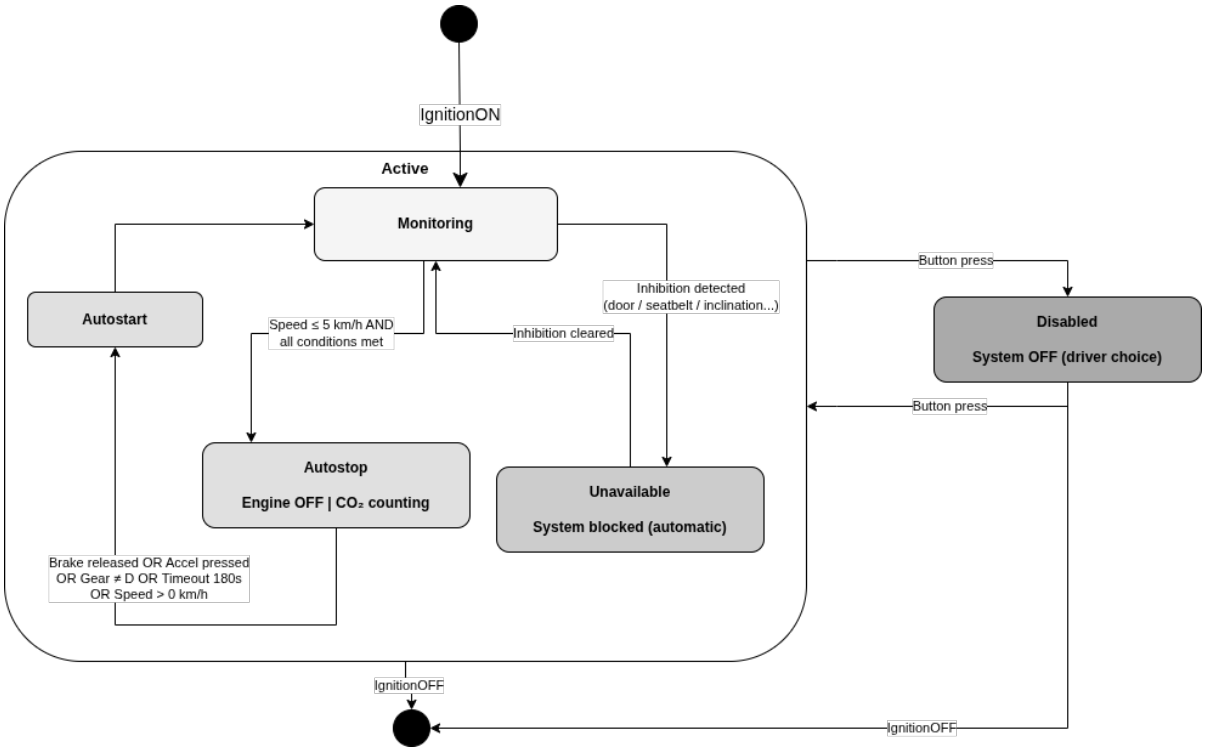


Figure 4: ESS State Machine Diagram

5 Simplified Requirements Diagram

This section presents the requirements diagram of the Start/Stop system.

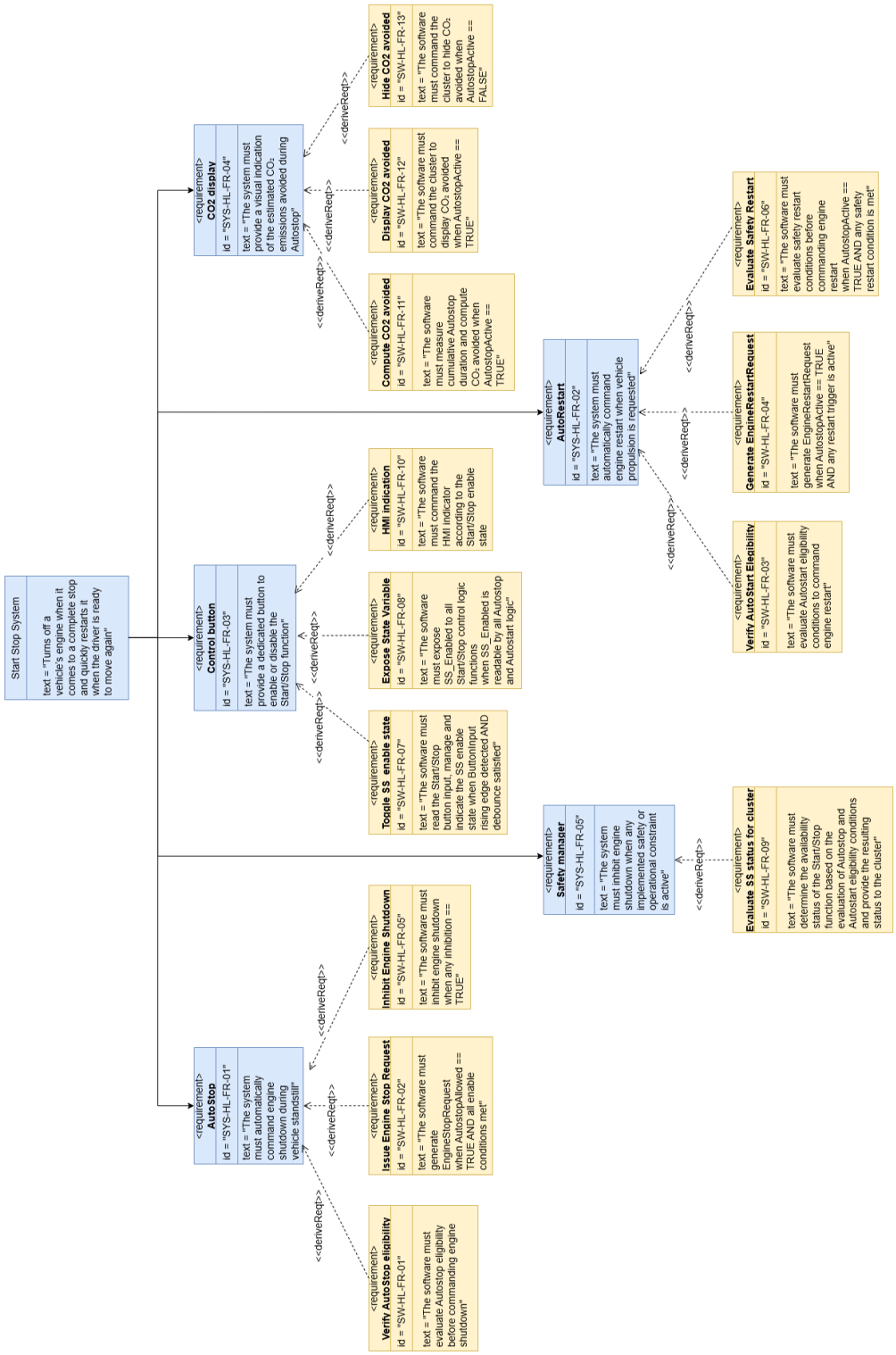


Figure 5: ESS Requirements Diagram

6 Functional Requirements

This section presents the functional requirements of the Engine Start-Stop System, organized in a four-level hierarchy. System-level requirements define what the system must do from an external perspective, while software-level requirements specify the internal behavior of the ESS software module. Each requirement is uniquely identified, prioritized, and traceable to its parent requirement at the level above.

6.1 System Requirements

6.1.1 High Level (SYS-HL)

Code	Action	Description	Priority
SYS-HL-FR-01	The system must automatically command engine shutdown during vehicle standstill.	The Start/Stop system must reduce fuel consumption and emissions by stopping the engine when the vehicle is stationary and operating conditions allow the function.	High
SYS-HL-FR-02	The system must automatically command engine restart when vehicle propulsion is requested.	The system must restart the engine to allow the driver to resume vehicle motion when propulsion demand is detected.	High
SYS-HL-FR-03	The system must provide a dedicated button to enable or disable the Start/Stop function.	The driver must be able to manually enable or disable the Start/Stop functionality through a dedicated control interface.	High
SYS-HL-FR-04	The system must provide a visual indication of the estimated CO_2 emissions avoided during Autostop.	The system must inform the driver about the environmental benefit associated with the Start/Stop operation.	Low
SYS-HL-FR-05	The system must inhibit engine shutdown when any implemented safety or operational constraint is active.	The Start/Stop function must be disabled when vehicle or driver safety conditions do not allow safe engine shutdown.	High

6.1.2 Low Level (SYS-LL)

Code	Action	Condition	Traceability
SYS-LL-FR-01	The system must automatically shut down the engine.	Start/Stop is enabled AND gear selector is in Drive AND brake pedal is pressed AND vehicle speed is at or below standstill threshold.	SYS-HL-FR-01
SYS-LL-FR-02	The system must automatically restart the engine.	Start/Stop is enabled AND Autostop is active AND (brake pedal is released OR gear selector is moved out of Drive OR accelerator pedal is pressed).	SYS-HL-FR-02
SYS-LL-FR-03	The system must toggle the Start/Stop function state.	Start/Stop Button rising edge is detected AND debounce condition is satisfied.	SYS-HL-FR-03
SYS-LL-FR-04	The system must provide visual indication of the state transitions (Cluster).	Start/Stop enable state transitions between enabled and disabled.	SYS-HL-FR-03
SYS-LL-FR-05	The system must provide visual indication of the activation status (LED).	Start/Stop function is disabled.	SYS-HL-FR-03
SYS-LL-FR-06	The system must set the Start/Stop status to Unavailable and inform the driver that the function cannot operate.	Start/Stop function is unavailable.	SYS-HL-FR-05
SYS-LL-FR-07	The system must display the estimated CO_2 avoided on the cluster.	Autostop is active.	SYS-HL-FR-04
SYS-LL-FR-08	The system must stop displaying CO_2 avoided.	Autostop transitions from active to inactive.	SYS-HL-FR-04
SYS-LL-FR-09	The system must compute CO_2 avoided as a function of Autostop duration and WLTP reference factor.	Autostop is active. CO2_AVOIDED = TIME_AUTO_STOP × WLTP_IDLE_FACTOR.	SYS-HL-FR-04

6.2 Software Requirements

6.2.1 High Level (SW-HL)

Code	Action	Condition	Traceability
SW-HL-FR-01	The software must evaluate Autostop eligibility before commanding engine shutdown.	All AutostopAllowed conditions evaluated each 100 ms cycle.	SYS-LL-FR-01, SYS-LL-FR-06
SW-HL-FR-02	The software must generate EngineStopRequest.	AutostopAllowed == TRUE AND all enable conditions met.	SYS-LL-FR-01, SYS-LL-FR-06
SW-HL-FR-03	The software must evaluate Autostart eligibility conditions to command engine restart.	All AutostartAllowed conditions evaluated each 100 ms cycle.	SYS-LL-FR-02, SYS-LL-FR-06
SW-HL-FR-04	The software must generate EngineRestartRequest.	AutostopActive == TRUE AND any restart trigger is active.	SYS-LL-FR-02, SYS-LL-FR-06
SW-HL-FR-05	The software must inhibit engine shutdown.	Any inhibition condition == TRUE.	SYS-LL-FR-01, SYS-LL-FR-06
SW-HL-FR-06	The software must evaluate safety restart conditions before commanding engine restart.	AutostopActive == TRUE AND any safety restart condition is met.	SYS-LL-FR-02, SYS-LL-FR-06
SW-HL-FR-07	The software must read the Start/Stop button input, manage and indicate the SS enable state.	ButtonInput rising edge detected AND debounce satisfied.	SYS-LL-FR-03, SYS-LL-FR-04, SYS-LL-FR-05
SW-HL-FR-08	The software must expose SS_Enabled to all Start/Stop control logic functions.	SS_Enabled is readable by all Autostop and Autostart logic.	SYS-LL-FR-03, SYS-LL-FR-05
SW-HL-FR-09	The software must determine the availability status of the Start/Stop function based on the evaluation of Autostop and Autostart eligibility conditions and provide the resulting status to the cluster.	—	SYS-LL-FR-06
SW-HL-FR-10	The software must command the HMI indicator according to the Start/Stop enable state.	—	SYS-LL-FR-04, SYS-LL-FR-05

Code	Action	Condition	Traceability
SW-HL-FR-11	The software must measure cumulative Autostop duration and compute CO_2 avoided.	AutostopActive == TRUE.	SYS-LL-FR-07, SYS-LL-FR-08, SYS-LL-FR-09
SW-HL-FR-12	The software must command the cluster to display CO_2 avoided.	AutostopActive == TRUE.	SYS-LL-FR-07, SYS-LL-FR-08, SYS-LL-FR-09
SW-HL-FR-13	The software must command the cluster to hide CO_2 avoided.	AutostopActive == FALSE.	SYS-LL-FR-07, SYS-LL-FR-08, SYS-LL-FR-09

6.2.2 Low Level (SW-LL)

Code	Category	Action	Condition	Traceability
SW-LL-FR-01	Engine Stop – Enable	The software must set EngineStopRequest = TRUE.	SS_Enabled == TRUE AND GearPosition == D AND BrakeStatus == PRESSED (BrakePedalPosition $\geq$ 5%) AND VehicleSpeed $\leq$ 5 km/h AND AutostopAllowed == TRUE.	SW-HL-FR-01, SW-HL-FR-02
SW-LL-FR-02	Engine Stop – Enable	The software must allow AutostopAllowed = TRUE.	SpeedThresholdReached == TRUE, where SpeedThresholdReached is set when VehicleSpeed > 8 km/h is detected in the current drive cycle.	SW-HL-FR-01, SW-HL-FR-02
SW-LL-FR-03	Engine Stop – Inhibition	The software must set AutostopAllowed = FALSE.	GearPosition != D.	SW-HL-FR-01, SW-HL-FR-05
SW-LL-FR-04	Engine Stop – Inhibition	The software must set AutostopAllowed = FALSE.	InclinationAngle > INCLINATION_THRESHOLD (15%).	SW-HL-FR-05
SW-LL-FR-05	Engine Stop – Inhibition	The software must set AutostopAllowed = FALSE.	DoorStatus != CLOSED.	SW-HL-FR-05
SW-LL-FR-06	Engine Stop – Inhibition	The software must set AutostopAllowed = FALSE.	SeatbeltStatus != FASTENED.	SW-HL-FR-05
SW-LL-FR-07	Engine Restart – Autostart	The software must set EngineRestartRequest = TRUE.	AutostopActive == TRUE AND AcceleratorStatus == PRESSED (AcceleratorPedalPosition > 5%).	SW-HL-FR-03, SW-HL-FR-04
SW-LL-FR-08	Engine Restart – Autostart	The software must set EngineRestartRequest = TRUE.	AutostopActive == TRUE AND BrakeStatus == RELEASED (BrakePedalPosition < 5%).	SW-HL-FR-03, SW-HL-FR-04
SW-LL-FR-09	Engine Restart – Autostart	The software must set EngineRestartRequest = TRUE.	AutostopActive == TRUE AND GearPosition != D.	SW-HL-FR-03, SW-HL-FR-04, SW-HL-FR-06

Code	Category	Action	Condition	Traceability
SW-LL-FR-10	Engine Restart – Autostart	The software must set EngineRestartRequest = TRUE.	AutostopActive == TRUE AND TIME_AUTO_STOP ≥ 180 000 ms.	SW-HL-FR-06
SW-LL-FR-11	Enable / Disable	The software must set SS_Enabled = TRUE.	Start_Ignition == TRUE.	SW-HL-FR-07, SW-HL-FR-08
SW-LL-FR-12	Enable / Disable	The software must toggle SS_Enabled.	ButtonInput rising edge detected AND ButtonInput stable ≥ T_debounce ms (50 ms).	SW-HL-FR-07, SW-HL-FR-08
SW-LL-FR-13	Enable / Disable	The software must reject button input.	ButtonInput stable < T_debounce ms (50 ms).	SW-HL-FR-07, SW-HL-FR-08
SW-LL-FR-14	Enable / Disable	The software must set AutostopAllowed = FALSE AND AutostartAllowed = FALSE.	SS_Enabled == FALSE.	SW-HL-FR-07, SW-HL-FR-08
SW-LL-FR-15	Cluster	The software must send a display message for 3 s to the cluster.	SS_Enabled != SS_Enabled_prev.	SW-HL-FR-07
SW-LL-FR-16	Unavailable	The software must set SS_Unavailable = TRUE.	AutoStopAllowed == FALSE OR AutoStartAllowed == FALSE AND SS_Enabled == TRUE.	SW-HL-FR-09
SW-LL-FR-17	Cluster	The software must command the instrument cluster to display the Start/Stop Unavailable indication.	SS_Unavailable == TRUE.	SW-HL-FR-09
SW-LL-FR-18	Cluster	The software must command the instrument cluster to clear the Start/Stop Unavailable indication.	SS_Unavailable == FALSE.	SW-HL-FR-09
SW-LL-FR-19	Cluster	The software must display AUTOSTOP_ACTIVE indicator in the cluster.	AutostopActive == TRUE.	SW-HL-FR-07
SW-LL-FR-20	HMI	The software must set the Start/Stop HMI LED to OFF.	SS_Enabled == TRUE.	SW-HL-FR-10
SW-LL-FR-21	HMI	The software must set the Start/Stop HMI LED to ON.	SS_Enabled == FALSE.	SW-HL-FR-10
SW-LL-FR-22	HMI	The software must send a LED update command to the HMI LED.	SS_Enabled != SS_Enabled_prev.	SW-HL-FR-10
SW-LL-FR-23	CO_2 Estimation	The software must increment TIME_AUTO_STOP by 100 ms every control cycle.	AutostopActive == TRUE.	SW-HL-FR-11
SW-LL-FR-24	CO_2 Estimation	The software must reset TIME_AUTO_STOP = 0.	IgnitionOFF == TRUE.	SW-HL-FR-11

Code	Category	Action	Condition	Traceability
SW-LL-FR-25	CO_2 Estimation	The software must compute CO2_AVOIDED = TIME_AUTO_STOP × WLTP_IDLE_FACTOR.	AutostopActive == TRUE.	SW-HL-FR-11
SW-LL-FR-26	CO_2 Estimation	The software must command the cluster to display CO2_AVOIDED.	AutostopActive == TRUE.	SW-HL-FR-12
SW-LL-FR-27	CO_2 Estimation	The software must command the cluster to hide CO2_AVOIDED.	AutostopActive == FALSE.	SW-HL-FR-13

7 Non-Functional Requirements

This section defines the non-functional requirements of the Engine Start-Stop System, covering performance, timing, and safety constraints. These requirements do not describe specific system behaviors but establish the quality attributes and operational boundaries within which the ESS must function. Compliance with these requirements is essential to ensure system safety, driver experience, and regulatory alignment with ISO 26262 and WLTP standards.

7.1 System Requirements

7.1.1 High Level (SYS-HL)

Code	Action	Traceability
SYS-HL-NFR-01	The system must be developed in accordance with ISO 26262. HARA must be performed and appropriate safety mechanisms implemented to ensure safe engine shutdown and restart under all operating conditions.	SYS-HL-FR-05
SYS-HL-NFR-02	The system must reduce CO_2 emissions by at least 5% under WLTP urban cycle compared to baseline operation without Start/Stop.	SYS-HL-FR-01, SYS-HL-FR-02, SYS-HL-FR-04
SYS-HL-NFR-03	The system must reduce fuel consumption by at least 5% under WLTP urban cycle compared to baseline operation without Start/Stop.	SYS-HL-FR-01

7.1.2 Low Level (SYS-LL)

Code	Action	Traceability
SYS-LL-NFR-01	The engine stop/restart response time must be less than defined threshold.	SYS-LL-FR-01, SYS-LL-FR-02
SYS-LL-NFR-02	The system must provide clear and timely visual feedback to the driver regarding the operational status of the Start/Stop feature.	SYS-LL-FR-04, SYS-LL-FR-05, SYS-LL-FR-06, SYS-LL-FR-07, SYS-LL-FR-08

7.2 Software Requirements

7.2.1 High Level (SW-HL)

Code	Action	Traceability
SW-HL-NFR-01	The execution time of the decision algorithm must not exceed 20 ms per cycle.	SW-HL-FR-01, SW-HL-FR-03
SW-HL-NFR-02	The software must update the Start/Stop decision every 100 ms.	SW-HL-FR-01, SW-HL-FR-03

7.2.2 Low Level (SW-LL)

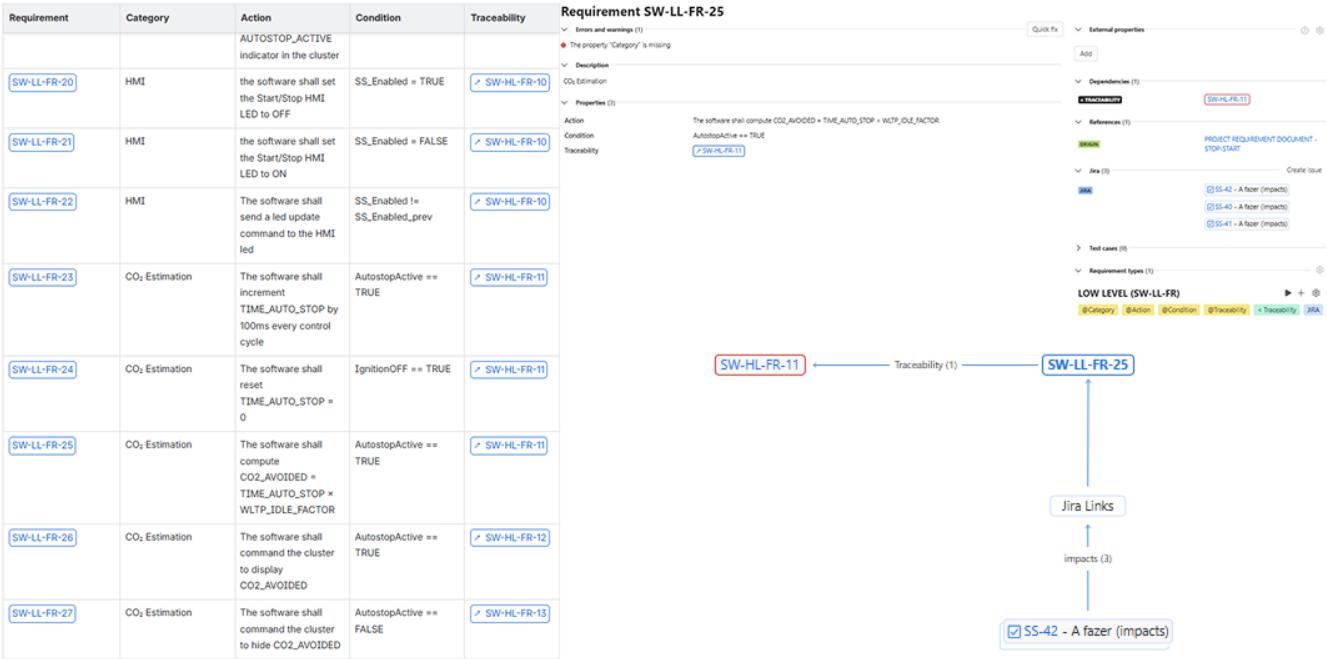
Code	Action	Traceability
SW-LL-NFR-01	The software must prioritise safety-related conditions over fuel-saving or cost-optimisation strategies. If any safety constraint is active, engine shutdown must be inhibited regardless of efficiency benefits.	SW-LL-FR-04, SW-LL-FR-05, SW-LL-FR-06
SW-LL-NFR-02	The software must detect a stable button press and update SS_Enabled within T_RESPONSE_MAX ms (100 ms).	SW-LL-FR-13

8 Requirement Traceability

For management and tracking purposes in Jira, the Yogi Requirements plug-in is being used. This tool enables the team to formally register and organize system requirements within Jira in a structured way.

With Yogi Requirements, each requirement can be created as a distinct item and assigned a unique key or identifier. This key allows the requirement to be easily referenced and tracked throughout the development lifecycle. Once the requirements are defined, they can be linked to Jira issues (such as tasks, stories, or bugs) that are created to implement or address those requirements. This linkage provides traceability, making it possible to track which development activities correspond to each requirement and to monitor the progress of implementation directly from Jira.

In addition, the project uses the integration between Jira and GitHub. This integration allows to connect Jira issues with code repositories and development activities in GitHub. As a result, actions such as commits, branches, and pull requests can be associated with specific Jira issues. This improves collaboration and transparency, since all members can see how code changes relate to the tasks and requirements being tracked in Jira.



Requirement	Category	Action	Condition	Traceability
		AUTOSTOP_ACTIVE indicator in the cluster		
SW-LL-FR-20	HMI	the software shall set the Start/Stop HMI LED to OFF	SS_Enabled = TRUE	SW-HL-FR-10
SW-LL-FR-21	HMI	the software shall set the Start/Stop HMI LED to ON	SS_Enabled = FALSE	SW-HL-FR-10
SW-LL-FR-22	HMI	The software shall send a led update command to the HMI led	SS_Enabled != SS_Enabled_prev	SW-HL-FR-10
SW-LL-FR-23	CO_2 Estimation	The software shall increment TIME_AUTO_STOP by 100ms every control cycle	AutostopActive == TRUE	SW-HL-FR-11
SW-LL-FR-24	CO_2 Estimation	The software shall reset TIME_AUTO_STOP = 0	IgnitionOFF == TRUE	SW-HL-FR-11
SW-LL-FR-25	CO_2 Estimation	The software shall compute CO2_AVOIDED = TIME_AUTO_STOP × WLTP_IDLE_FACTOR	AutostopActive == TRUE	SW-HL-FR-11
SW-LL-FR-26	CO_2 Estimation	The software shall command the cluster to display CO2_AVOIDED	AutostopActive == TRUE	SW-HL-FR-12
SW-LL-FR-27	CO_2 Estimation	The software shall command the cluster to hide CO2_AVOIDED	AutostopActive == FALSE	SW-HL-FR-13

Figure 6: Requirement Yogi Management

9 Traceability Matrix

This section presents the full traceability mapping between requirement levels, following the derivation chain from System High-Level down to Software Low-Level. Each table reads as: the requirement on the left column is the parent, and the requirements on the right column are directly derived from it.

This matrix ensures that every high-level stakeholder need is covered by at least one lower-level requirement, and that no low-level requirement exists without a justified origin. It also supports verification activities by enabling bidirectional traceability — from implementation back to the original system need.

9.1 System High-Level to System Low-Level

System HL Req.	Derived System LL Requirements
SYS-HL-FR-01	SYS-LL-FR-01
SYS-HL-FR-02	SYS-LL-FR-02
SYS-HL-FR-03	SYS-LL-FR-03, SYS-LL-FR-04, SYS-LL-FR-05
SYS-HL-FR-04	SYS-LL-FR-07, SYS-LL-FR-08, SYS-LL-FR-09
SYS-HL-FR-05	SYS-LL-FR-06

9.2 System Low-Level to Software High-Level

System LL Req.	Derived Software HL Requirements
SYS-LL-FR-01	SW-HL-FR-01, SW-HL-FR-02
SYS-LL-FR-02	SW-HL-FR-03, SW-HL-FR-04, SW-HL-FR-06
SYS-LL-FR-03	SW-HL-FR-07, SW-HL-FR-08
SYS-LL-FR-04	SW-HL-FR-07, SW-HL-FR-10
SYS-LL-FR-05	SW-HL-FR-07, SW-HL-FR-10
SYS-LL-FR-06	SW-HL-FR-01, SW-HL-FR-02, SW-HL-FR-03, SW-HL-FR-04, SW-HL-FR-05, SW-HL-FR-06, SW-HL-FR-09
SYS-LL-FR-07	SW-HL-FR-11, SW-HL-FR-12
SYS-LL-FR-08	SW-HL-FR-11, SW-HL-FR-13
SYS-LL-FR-09	SW-HL-FR-11

9.3 Software High-Level to Software Low-Level

Software HL Req.	Derived Software LL Requirements
SW-HL-FR-01	SW-LL-FR-01, SW-LL-FR-02, SW-LL-FR-03
SW-HL-FR-02	SW-LL-FR-01, SW-LL-FR-02
SW-HL-FR-03	SW-LL-FR-07, SW-LL-FR-08, SW-LL-FR-09
SW-HL-FR-04	SW-LL-FR-07, SW-LL-FR-08, SW-LL-FR-09
SW-HL-FR-05	SW-LL-FR-03, SW-LL-FR-04, SW-LL-FR-05, SW-LL-FR-06
SW-HL-FR-06	SW-LL-FR-09, SW-LL-FR-10
SW-HL-FR-07	SW-LL-FR-11, SW-LL-FR-12, SW-LL-FR-13, SW-LL-FR-14, SW-LL-FR-15, SW-LL-FR-19
SW-HL-FR-08	SW-LL-FR-11, SW-LL-FR-12, SW-LL-FR-13, SW-LL-FR-14
SW-HL-FR-09	SW-LL-FR-16, SW-LL-FR-17, SW-LL-FR-18
SW-HL-FR-10	SW-LL-FR-20, SW-LL-FR-21, SW-LL-FR-22
SW-HL-FR-11	SW-LL-FR-23, SW-LL-FR-24, SW-LL-FR-25
SW-HL-FR-12	SW-LL-FR-26
SW-HL-FR-13	SW-LL-FR-27

9.4 Non-Functional Requirements to Functional Requirements

Non-Functional Requirement	Functional Requirement
SYS-HL-NFR-01	SYS-HL-FR-05
SYS-HL-NFR-02	SYS-HL-FR-01, SYS-HL-FR-02, SYS-HL-FR-04
SYS-HL-NFR-03	SYS-HL-FR-01
SYS-LL-NFR-01	SYS-LL-FR-01, SYS-LL-FR-02
SYS-LL-NFR-02	SYS-LL-FR-04, SYS-LL-FR-05, SYS-LL-FR-06, SYS-LL-FR-07, SYS-LL-FR-08
SW-HL-NFR-01	SW-HL-FR-01, SW-HL-FR-03
SW-HL-NFR-02	SW-HL-FR-01, SW-HL-FR-03
SW-LL-NFR-01	SW-LL-FR-04, SW-LL-FR-05, SW-LL-FR-06
SW-LL-NFR-02	SW-LL-FR-13

10 Appendix

10.1 Use Case Specifications

This appendix presents the detailed specification of the main use cases of the Start/Stop system. Each use case describes the actors involved, the system behavior, and the main and alternative flows.

10.1.1 Use Case 1 – Execute Auto Stop

Name	Execute Auto Stop
Description	This use case describes the automatic engine shutdown performed by the Start/Stop system when the vehicle reaches standstill and all Autostop eligibility conditions are satisfied. During this condition the system commands engine shutdown, activates the AUTOSTOP indicator, and starts the Estimate CO_2 avoided routine.
Actors	Driver, Powertrain ECU, Instrument Cluster / HMI
Preconditions	Start/Stop system initialized; Start/Stop function enabled; vehicle ignition ON; Autostop eligibility conditions satisfied.
Postconditions	EngineStopRequest is generated and sent to the Powertrain ECU. The engine enters Autostop condition and the AUTOSTOP indicator is displayed.
Main Flow [FP]	1. Vehicle reaches standstill. 2. The Start/Stop system evaluates Autostop eligibility. 3. The system verifies that AutostopAllowed is TRUE. 4. The system generates EngineStopRequest. 5. Powertrain ECU performs engine shutdown. 6. AUTOSTOP ACTIVE indicator is displayed. 7. Estimate CO_2 avoided routine starts.
Alternative Flow [AF]	1. Autostop eligibility conditions are not satisfied. 2. EngineStopRequest is not generated. 3. The engine continues running.

10.1.2 Use Case 2 – Execute Auto Start

Name	Execute Auto Start
Description	This use case describes the automatic restart of the engine when propulsion demand or safety conditions require the engine to restart during Autostop condition.
Actors	Driver, Powertrain ECU
Preconditions	Engine is in Autostop condition; Start/Stop enabled; ignition ON.
Postconditions	EngineRestartRequest is generated and sent to the Powertrain ECU. Engine returns to running state.
Main Flow [FP]	1. Restart trigger is detected. 2. System evaluates Autostart eligibility. 3. System generates EngineRestartRequest. 4. Powertrain ECU performs engine restart.
Alternative Flow [AF]	1. Restart conditions are not satisfied. 2. EngineRestartRequest is not generated. 3. Engine remains in Autostop condition.

10.1.3 Use Case 3 – Enable / Disable Start Stop

Name	Enable / Disable Start Stop (Button)
Description	This use case describes the interaction of the driver with the Start/Stop control button used to enable or disable the Start/Stop functionality.
Actors	Driver, Instrument Cluster / HMI
Preconditions	Vehicle ignition ON; Start/Stop system initialized.
Postconditions	Start/Stop enable state updated and displayed to the driver.
Main Flow [FP]	1. Driver presses the Start/Stop button. 2. System reads button input. 3. Debounce logic validates input. 4. System toggles Start/Stop enable state. 5. HMI displays updated Start/Stop status.
Alternative Flow [AF]	1. Button press does not satisfy debounce condition. 2. Input is ignored. 3. System state remains unchanged.

10.1.4 Use Case 4 – Display AUTOSTOP ACTIVE Indicator

Name	Display AUTOSTOP ACTIVE Indicator
Description	Displays a visual indication informing the driver that the engine has been automatically stopped by the Start/Stop system.
Actors	Instrument Cluster / HMI
Preconditions	Engine is in Autostop condition.
Postconditions	AUTOSTOP ACTIVE indicator is visible in the instrument cluster.
Main Flow	1. Start/Stop system detects Autostop state. 2. System sends indicator command to HMI. 3. HMI displays AUTOSTOP ACTIVE indicator.
Alternative Flow	Indicator cannot be displayed due to HMI failure.

10.1.5 Use Case 5 – Estimate CO_2 avoided

Name	Estimate CO_2 avoided
Description	Estimates the amount of CO_2 emissions avoided during Autostop condition based on cumulative idle time and a WLTP reference factor.
Actors	Instrument Cluster / HMI
Preconditions	Autostop condition active; time accumulation running.
Postconditions	Estimated CO_2 avoided value calculated and available for display.
Main Flow	1. System detects Autostop state. 2. Autostop duration is accumulated. 3. CO_2 avoided value is computed. 4. Value is stored for visualization.
Alternative Flow	Autostop ends before calculation update.

10.1.6 Use Case 6 – Display CO_2 Avoided During AutoStop

Name	Display CO_2 Avoided During AutoStop
Description	Displays the estimated CO_2 emissions avoided while the vehicle remains in Autostop condition.
Actors	Instrument Cluster / HMI
Preconditions	Autostop condition active; CO_2 avoided value computed.
Postconditions	CO_2 avoided information shown to the driver.
Main Flow	1. System retrieves CO_2 avoided value. 2. System sends display command to HMI. 3. HMI displays CO_2 information.
Alternative Flow	Autostop ends and CO_2 information is removed from display.

10.1.7 Use Case 7 – Evaluate AutoStop Eligibility

Name	Evaluate AutoStop Eligibility
Description	Evaluates whether all required conditions are satisfied to allow automatic engine shutdown.
Actors	Driver
Preconditions	Start/Stop system initialized; ignition ON.
Postconditions	AutostopAllowed state updated.
Main Flow	1. System reads vehicle operating signals. 2. System evaluates Autostop enabling conditions. 3. System verifies absence of inhibition conditions. 4. AutostopAllowed state updated.
Alternative Flow	Autostop inhibited due to safety condition.

10.1.8 Use Case 8 – Evaluate Autostart Eligibility

Name	Evaluate Autostart Eligibility
Description	Evaluates whether the engine should be restarted due to propulsion demand or safety restart conditions.
Actors	Driver
Preconditions	Engine currently in Autostop state.
Postconditions	Restart eligibility state updated.
Main Flow	1. System reads restart trigger inputs. 2. System evaluates safety restart conditions. 3. System determines restart eligibility.
Alternative Flow	No valid restart trigger detected.

10.1.9 Use Case 9 – Generate EngineStopRequest

Name	Generate EngineStopRequest
Description	Generates the engine stop request command sent to the Powertrain ECU.
Actors	Powertrain ECU
Preconditions	AutostopAllowed == TRUE.
Postconditions	EngineStopRequest transmitted to ECU.
Main Flow	System generates and sends EngineStopRequest to Powertrain ECU.
Alternative Flow	Shutdown request not issued if conditions are invalid.

10.1.10 Use Case 10 – Generate EngineRestartRequest

Name	Generate EngineRestartRequest
Description	Generates the engine restart request sent to the Powertrain ECU.
Actors	Powertrain ECU
Preconditions	Restart conditions detected.
Postconditions	EngineRestartRequest transmitted to ECU.
Main Flow	System generates restart command and sends it to ECU.
Alternative Flow	Restart request not issued if conditions are invalid.

10.1.11 Use Case 11 – Indicate StartStop Enabled State

Name	Indicate StartStop Enabled State
Description	Displays the current Start/Stop enabled state to the driver through the HMI.
Actors	Instrument Cluster / HMI
Preconditions	Start/Stop state variable available.
Postconditions	Driver receives updated Start/Stop status indication.
Main Flow	System sends state indication command to HMI.
Alternative Flow	HMI unavailable.

10.1.12 Use Case 12 – Initialize StartStop

Name	Initialize StartStop
Description	Initializes Start/Stop system at ignition cycle start.
Actors	Ignition System
Preconditions	Ignition ON event detected.
Postconditions	System variables initialized and Start/Stop system ready.
Main Flow	System initializes internal states and enables operation.
Alternative Flow	Initialization fails and system remains unavailable.

10.1.13 Use Case 13 – Initialize Time Auto Stop

Name	Initialize Time Auto Stop
Description	Initializes the accumulated Autostop time counter when ignition cycle starts.
Actors	Ignition System
Preconditions	Ignition ON event.
Postconditions	TIME_AUTO_STOP variable reset to zero.
Main Flow	System resets Autostop timer variable.
Alternative Flow	Timer reset fails and system attempts reinitialization.

10.2 Traceability Matrix - GERSE

SYSTEM LOW-LEVEL X SYSTEM HIGH-LEVEL (FUNCTIONAL)					
SYS-LL-FR \ SYS-HL-FR	SYS-HL-FR-01	SYS-HL-FR-02	SYS-HL-FR-03	SYS-HL-FR-04	SYS-HL-FR-05
SYS-LL-FR-01	X				
SYS-LL-FR-02		X			
SYS-LL-FR-03			X		
SYS-LL-FR-04			X		
SYS-LL-FR-05			X		
SYS-LL-FR-06					X
SYS-LL-FR-07				X	
SYS-LL-FR-08				X	
SYS-LL-FR-09				X	

Figure 7: System (Functional) - Low Level X High Level

SOFTWARE HIGH-LEVEL X SYSTEM LOW-LEVEL (FUNCTIONAL)									
SW-HL-FR \ SYS-LL-FR	SYS-LL-FR-01	SYS-LL-FR-02	SYS-LL-FR-03	SYS-LL-FR-04	SYS-LL-FR-05	SYS-LL-FR-06	SYS-LL-FR-07	SYS-LL-FR-08	SYS-LL-FR-09
SW-HL-FR-01	X					X			
SW-HL-FR-02	X					X			
SW-HL-FR-03		X				X			
SW-HL-FR-04		X				X			
SW-HL-FR-05	X					X			
SW-HL-FR-06		X	X			X			
SW-HL-FR-07			X	X	X				
SW-HL-FR-08					X				
SW-HL-FR-09						X			
SW-HL-FR-10				X	X		X		
SW-HL-FR-11							X	X	X
SW-HL-FR-12							X	X	X
SW-HL-FR-13								X	X

Figure 8: Software X System (Functional) - Low Level X High Level

SOFTWARE LOW-LEVEL X SOFTWARE HIGH-LEVEL (FUNCTIONAL)													
SW-LL-FR \ SW-HL-FR	SW-HL-FR-01	SW-HL-FR-02	SW-HL-FR-03	SW-HL-FR-04	SW-HL-FR-05	SW-HL-FR-06	SW-HL-FR-07	SW-HL-FR-08	SW-HL-FR-09	SW-HL-FR-10	SW-HL-FR-11	SW-HL-FR-12	SW-HL-FR-13
SW-LL-FR-01	X	X											
SW-LL-FR-02	X	X											
SW-LL-FR-03	X				X								
SW-LL-FR-04					X								
SW-LL-FR-05					X								
SW-LL-FR-06					X								
SW-LL-FR-07			X	X									
SW-LL-FR-08			X	X									
SW-LL-FR-09			X	X		X							
SW-LL-FR-10						X							
SW-LL-FR-11							X	X					
SW-LL-FR-12							X	X					
SW-LL-FR-13							X	X					
SW-LL-FR-14							X	X					
SW-LL-FR-15							X						
SW-LL-FR-16									X				
SW-LL-FR-17									X				
SW-LL-FR-18									X				
SW-LL-FR-19							X						
SW-LL-FR-20										X			
SW-LL-FR-21										X			
SW-LL-FR-22										X			
SW-LL-FR-23											X		
SW-LL-FR-24											X		
SW-LL-FR-25											X		
SW-LL-FR-26												X	
SW-LL-FR-27													X

Figure 9: Software (Functional) - Low Level X High Level

FUNCTIONAL REQUIREMENTS X NON FUNCTIONAL REQUIREMENTS									
FUNCTIONAL \ NON-FUNCTIONAL	SW-HL-NFR-01	SW-HL-NFR-02	SW-LL-NFR-01	SW-LL-NFR-02	SYS-HL-NFR-01	SYS-HL-NFR-02	SYS-HL-NFR-03	SYS-LL-NFR-01	SYS-LL-NFR-02
SW-HL-FR-01	X	X							
SW-HL-FR-03	X	X							
SW-LL-FR-04			X						
SW-LL-FR-05			X						
SW-LL-FR-06			X						
SW-LL-FR-13				X					
SYS-HL-FR-01						X	X		
SYS-HL-FR-02						X			
SYS-HL-FR-04						X			
SYS-HL-FR-05					X				
SYS-LL-FR-01								X	
SYS-LL-FR-02								X	
SYS-LL-FR-04									X
SYS-LL-FR-05									X
SYS-LL-FR-06									X
SYS-LL-FR-07									X
SYS-LL-FR-08									X

Figure 10: Functional Requirements X Non Functional Requirements

11 References

The requirements were initially elicited through a review of scientific literature and original manufacturer documentation on engine Start/Stop systems. Not all requirements identified in these sources could be fully contemplated within the scope of this project due to time and resource constraints.

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